WORKED SOLUTION

STEP II 2001, Question 9

Mechanics • Quadratic Air Resistance

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Read the question before you go on. It is in the STEP Support Programme's question database:

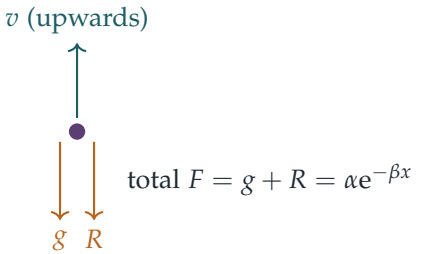
step.maths.org/questions/01-s2-q9

The key idea

The force is given as a function of *height*, not time, so the natural tool is the work–energy principle: the work done against F over the climb is exactly the kinetic energy lost. That links $\mathrm{e}^{-\beta x}$ to v^2, and the result shows that the air resistance depends only on the speed. Once we know the resistance law in terms of speed, the downward journey is a standard differential equation in v^2.

Solution

Model. The particle has unit mass and moves on a vertical line. On the way up, both gravity and air resistance act downwards, against the motion, so the total force F (magnitude $\alpha\mathrm{e}^{-\beta x}$) acts downwards. Take x upwards from the point of projection.



Energy expended

The energy expended in reaching height x is the work done against F:

$$\int_0^x \alpha\mathrm{e}^{-\beta s}\,\mathrm{d}s = \frac{\alpha}{\beta}\left(1-\mathrm{e}^{-\beta x}\right).$$

This energy comes from the particle's kinetic energy, so by the work–energy principle (unit mass)

$$\tfrac{1}{2}u^2 - \tfrac{1}{2}v^2 = \frac{\alpha}{\beta}\left(1 - \mathrm{e}^{-\beta x}\right). \qquad (*)$$

(Equivalently, Newton's second law in the form $v\frac{\mathrm{d}v}{\mathrm{d}x} = -\alpha\mathrm{e}^{-\beta x}$, integrated from $x = 0$, $v = u$, gives the same equation.)

The force in terms of speed

Multiply $(*)$ by β and rearrange:

$$\alpha - \alpha\mathrm{e}^{-\beta x} = \tfrac{1}{2}\beta u^2 - \tfrac{1}{2}\beta v^2 \quad \Longrightarrow \quad F = \alpha\mathrm{e}^{-\beta x} = \tfrac{1}{2}\beta v^2 + \alpha - \tfrac{1}{2}\beta u^2,$$

as required.

> **Ask yourself** Why must $\alpha = \frac{1}{2}\beta u^2 + g$? Nothing in the question seems to force it.

Split F into its two parts: $F = g + R$, where R is the air resistance. Then

$$R = F - g = \tfrac{1}{2}\beta v^2 + \left(\alpha - \tfrac{1}{2}\beta u^2 - g\right).$$

So the resistance is a function of the speed alone: a v^2 term plus a constant. Air resistance is caused by motion through the air, so it must vanish when the particle is momentarily at rest, at the highest point. Putting $v = 0$ there, $R = 0$, so $F = g$ (only gravity acts) and

$$\alpha - \tfrac{1}{2}\beta u^2 = g, \qquad \text{that is,} \qquad \alpha = \tfrac{1}{2}\beta u^2 + g.$$

Consequently the constant term is zero and the resistance law is simply

$$R = \tfrac{1}{2}\beta v^2.$$

(For the record, the highest point is where $\alpha\mathrm{e}^{-\beta x} = g$, at height $\frac{1}{\beta}\ln(\alpha/g)$.)

The downward journey

Assume the same resistance law $R = \frac{1}{2}\beta v^2$ holds whatever the direction of motion; on the way down it acts *upwards*, opposing the motion, while gravity still acts downwards. Let y be the distance below the highest point and v the downward speed, with $v = 0$ at $y = 0$. Newton's second law (unit mass), downwards:

$$v\frac{\mathrm{d}v}{\mathrm{d}y} = g - \tfrac{1}{2}\beta v^2.$$

Since $v\frac{\mathrm{d}v}{\mathrm{d}y} = \frac{1}{2}\frac{\mathrm{d}}{\mathrm{d}y}(v^2)$, put $w = v^2$:

$$\frac{\mathrm{d}w}{\mathrm{d}y} = 2g - \beta w, \qquad w = 0 \text{ at } y = 0.$$

This is linear with constant coefficients (or separate the variables): the solution is

$$w = \frac{2g}{\beta}\left(1 - \mathrm{e}^{-\beta y}\right),$$

which indeed satisfies $w(0) = 0$ and $w' = 2g\mathrm{e}^{-\beta y} = 2g - \beta w$. Therefore the air resistance at distance y below the highest point is

$$R = \tfrac{1}{2}\beta v^2 = g\left(1 - \mathrm{e}^{-\beta y}\right).$$

Checks

As $y \to \infty$, $R \to g$: the particle approaches terminal speed $\sqrt{2g/\beta}$, where resistance balances gravity. At $y = 0$, $R = 0$, as it should be at rest. On the way up, at $x = 0$ the formula gives $F = \alpha = g + \frac{1}{2}\beta u^2$, gravity plus the resistance at speed u. Numerically, with $g = 9.8$, $\beta = 0.1$, $u = 20$ (so $\alpha = 29.8$), stepping the equations of motion with a Runge–Kutta method reproduces $F = \alpha\mathrm{e}^{-\beta x}$ on the way up and $R = g(1 - \mathrm{e}^{-\beta y})$ on the way down to about 10^{-12}, and the particle peaks at height ≈ 11.12, matching $\frac{1}{\beta}\ln(\alpha/g)$.

Common slips

- Treating F as the air resistance alone: it is the *total* force, so gravity must be subtracted to get R.
- Keeping the same sign for the resistance on the way down: it always opposes the motion, so it now acts upwards and the equation becomes $v\frac{\mathrm{d}v}{\mathrm{d}y} = g - \frac{1}{2}\beta v^2$.
- Answering in terms of v, or of u and α; the question wants R in terms of y, g and β only.

Beyond STEP

Quadratic drag, $R \propto v^2$, is the realistic model for a fast-moving object in air, and this question shows a neat trick: in terms of distance, the equation for v^2 is *linear*, even though the equation for v in terms of time is not. Undergraduate mechanics also treats the upward-then-downward problem asymmetrically: the particle returns to its start with a speed less than u, and the descent takes longer than the ascent. Kleppner and Kolenkow, *An Introduction to Mechanics*, discusses velocity-dependent forces and terminal velocity in detail.

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